

Chapter 26: Advanced Electronic Materials and Emerging Device Technologies

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Abstract

The progress of modern electronics is inextricably tied to the discovery and development of new electronic materials and device technologies. It is a good semiconductor, easy to find, and has a well-established manufacturing infrastructure. Silicon has been the most important material in electronics. However, as the physical and performance limits of traditional silicon-based systems are reached, researchers are seeking to use new materials with better electrical, optical, magnetic and mechanical properties to enable next-generation devices. This chapter provides a broad overview of advanced electronic materials including two-dimensional (2D) materials, wide bandgap semiconductors, organic semiconductors, magnetic materials, and quantum materials. It considers new device types such as flexible electronics, spintronic devices, neuromorphic systems and quantum computing platforms. We discuss major fabrication methods, applications in various fields and major challenges such as scalability, stability and integration. The spotlight is on sustainable materials and magnetic systems without rare earths, especially manganese-based compounds such as MnBi and MnAl. The chapter ends with some thoughts on how advanced materials might be used in next generation electronic systems. It highlights the potential impact of these materials on technological revolution and sustainable development.

Keywords: 2D Materials, Thin films, Nanoparticles, Spintronics.

1. Introduction

Advances in materials science and engineering continue to propel the evolution of electronic gadgets. Discovering new materials and refining production techniques have enabled the development of every new technology, from vacuum tubes to transistors to integrated circuits. Silicon has long been the most essential material in electronics due to its excellent semiconductor properties, ease of availability, and well-established production infrastructure [1]. But as devices scale down to nanometer sizes, silicon-based technologies face major challenges such as increased leakage currents, heat dissipation problems and quantum mechanical effects. These problems are tackled by researchers looking for new electronic materials that can have better performance characteristics such as high carrier mobility, tunable bandgaps, mechanical flexibility and novel quantum properties. These materials are not only helping to extend Moore's law, but are opening up completely new ways of doing things in electronics, such as flexible devices, wearable technologies, spin-based computing and quantum information processing [2-4].

The purpose of this chapter is to provide a broad view of advanced electronic materials and of the new device technologies enabled by them. It includes the different kinds of materials, how to make them, how to build devices, how to use them, the problems they face and where they might go in the future.

2. Advanced Electronic Materials

2.1 Two-Dimensional (2D) Materials

Modern semiconductor technology is extending the domains of electronics, sensors, and energy technology, but it still poses a considerable challenge for two-dimensional (2D) materials. Despite substantial advances in traditional 2D materials, the intrinsic features of structure and electrical

configuration continue to limit device technology and integration compatibility in practical applications [1]. Updates and developments in the 2D material families are crucial for the next generation of multifunctional devices and nanostructured electronics. Two-dimensional materials are one or two atoms thick layers with amazing physical and electrical properties. The most famous example is graphene, a single sheet of carbon atoms arranged in a hexagonal lattice. Other notable 2D materials include transition metal dichalcogenides (TMDs), such as MoS₂ and WS₂, and insulating materials, such as hexagonal boron nitride. These materials are perfect for the electronics of the future because of their high carrier mobility, mechanical flexibility and large surface area. 2D materials can be tuned to have different electrical properties by changing the layer thickness, tension and chemical functionalization. 2D materials have been used in high-performance field-effect transistors (FETs), photodetectors, sensors, flexible electronic devices, etc [5-9].

2.3 Magnetic Materials and Spintronic Materials

Spintronic devices rely on magnetic materials for controlling and manipulating electron spin. Various categories of magnets, including ferromagnets with significant spontaneous magnetization and antiferromagnets with antiparallel spin alignments, provide distinct properties. Due to their superior magnetic properties, NdFeB magnets currently control the market. However, these magnets are largely reliant on the availability of rare earth elements (REEs). For the production of high-performance permanent magnets utilized in almost all contemporary technologies, rare earth elements (REEs), particularly Nd, Dy, and Tb, are in great demand [10]. However, China contributes the majority of the world's highly concentrated REE output. This concentration creates economic concerns, such as price volatility and supply chain risks. High-performance magnets are therefore exceedingly expensive to employ commercially due to the high cost of rare earth materials [10]. Due to the limited global supply of rare-earth elements and the rising demand for permanent magnet-based technology, the necessity for rare-earth-free (REF) magnets has garnered considerable attention [10]. Iron-based REF magnets, especially ferrites, are among the potential substitutes that are affordable and produced from easily accessible raw materials; nevertheless, their low maximal energy product (BH) limits their magnetic performance [10]. Although promising candidates like α -Fe₁₆N₂ and tetrataenite (FeNi) have surfaced, difficulties in creating bulk magnets or synthesizing polycrystalline materials have impeded their advancement. For consumer applications and moderate-performance motors, Mn-based magnets are a good substitute for REE- and Co-based magnets due to their simpler synthesis processes and lower cost [10]. Compared to REE-based magnets, Mn-based compounds have shown potential magnetic properties, ease of manufacture, and abundance in the Earth's crust at a far lower cost. Researchers have been particularly interested in manganese alloys, such as MnAl, MnGa, and MnBi, due to their strong magnetic anisotropy as possible rare-earth free permanent magnet candidates [10-13].

3. Fabrication Techniques

3.1 Thin Film Deposition Techniques

Thin films are layers of material that are only a few nanometers to several micrometres thick. They are widely used in electronics, optics, energy generation, sensing technologies, and many other applications. High performance and useful devices have been made possible by the distinct physical and chemical characteristics that result from materials' nanoscale structure [14]. As a result, there has been a growing interest in the various methods of thin-film deposition. Thermal evaporation, sputtering, chemical vapor

deposition (CVD), atomic layer deposition (ALD), molecular beam epitaxy (MBE), chemical bath deposition (CBD), and plasma-enhanced chemical vapor deposition (PECVD) are the most widely utilized deposition techniques in research and industrial applications [14].

Sputtering:

Sputtering is a versatile physical vapor deposition technique for coating thin films on surfaces. There are three sputtering processes, direct current (DC), radiofrequency (RF) and magnetron sputtering [14]. Sputtering is a physical vapor deposition process that utilizes high-energy ion bombardment to knock atoms off a target surface. The main steps of sputtering are: (1) The material under study is placed in a vacuum chamber; (2) Ion Bombardment: high energy ions are accelerated towards the material to be treated; (3) Target Atom Ejection: atoms are ejected from the target material when high energy ions collide with the target surface; (4) Plasma Formation: the ejected atoms form a plasma in the vacuum chamber; (5) Deposition on Substrate: The substrate is placed close to the target, and the ejected atoms condense on its surface to produce the necessary film properties, (6) Control and Monitoring: The process is carefully controlled, frequently utilizing techniques like magnetron sputtering or reactive sputtering, and monitored to produce the required film properties [14]. Figure 1 shows a simple illustration of the DC/RF sputtering process.

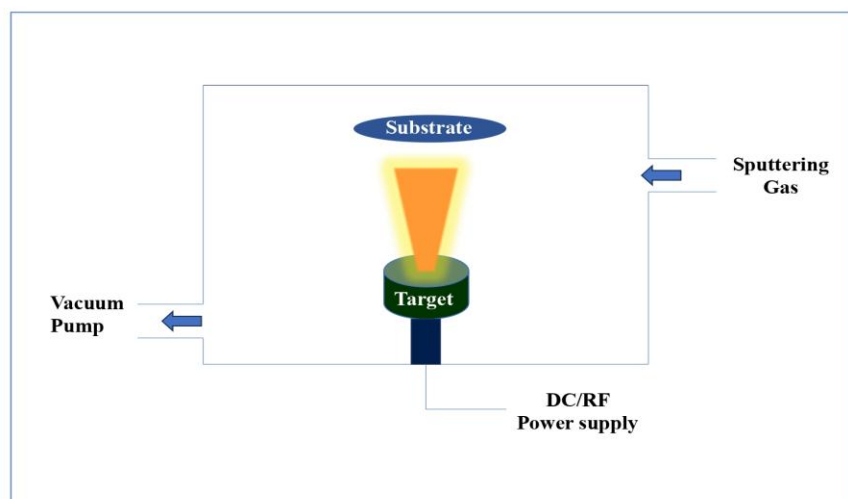


Fig.1. Simple schematic of a DC/RF sputtering system.

Evaporation Techniques:

Thermal evaporation [14] is a common technique to fabricate thin films in a number of commercial and scientific applications. This technique is based on thermal vaporization, which is the process of heating a substance to a high temperature to produce vapor and then condensing the vapor on a substrate to produce a thin layer [14]. The theory of thermal evaporation is founded on some fundamental equations of thermodynamics and molecular transport [14].

Chemical Vapor Deposition (CVD):

CVD is a widely known technique, which employs chemical reactions with gaseous precursors to generate thin films and coatings on surfaces. It does this by introducing volatile precursors into a reaction

chamber, where they chemically react to deposit a solid material on a substrate [15]. This very versatile process [14] can be used to manufacture a wide variety of materials including metals, semiconductors and insulators. A gaseous precursor is introduced into a reaction chamber and the substrate is exposed to it in a conventional CVD procedure [15]. These precursors are reacted with heat or plasma on the surface of the substrate and form a thin layer [15]. The deposition process is controlled precisely in terms of temperature, pressure and gas flow rates in order to produce the desired film characteristics as for thickness, homogeneity and composition [14].

Molecular Beam Epitaxy (MBE):

MBE is a very controlled ultrahigh vacuum (UHV) deposition method commonly used to create high purity crystalline thin films, particularly for semiconductor materials [16]. The basic principle of MBE is the controlled, stepwise deposition of atoms or molecules onto a heated substrate to form a crystalline structure in a layer-by-layer fashion [16]. Due to the extremely low deposition rates as compared to other deposition methods [14], MBE provides an unprecedented control over the layer thickness, composition and crystal quality. Figure 2 shows the schematic of a Molecular beam epitaxy chamber.

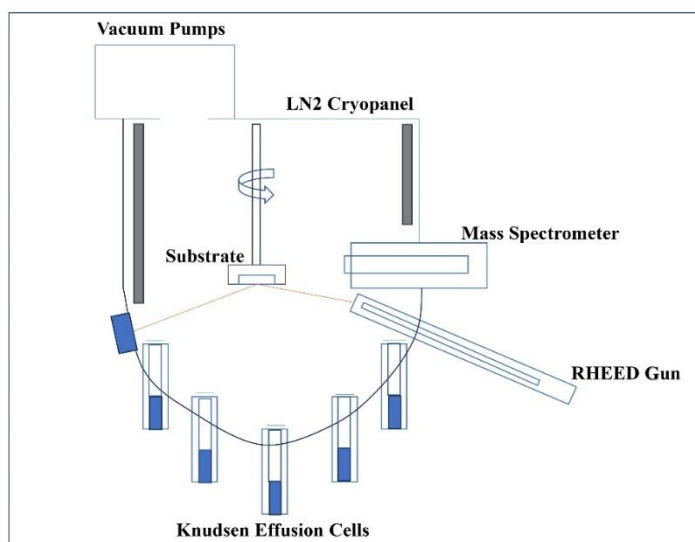


Fig.2. Schematic of a typical molecular beam epitaxy chamber.

3.2 Nanofabrication and Lithography

Lithography is critical in semiconductor manufacturing, allowing the development of smaller, more powerful electronic devices. Common lithographic techniques include photolithography, electron-beam lithography, nanoimprint lithography, and X-ray lithography [17]. Nanofabrication and lithography work together to build enhanced integrated circuits, microelectromechanical systems (MEMS), nano sensors, and next-generation electronic devices, accelerating advancements in current nanotechnology and semiconductor production [17].

4. Emerging Device Technologies

4.1 Spintronic Devices

One of the developing disciplines for next-generation nanoelectronic devices to lower their power consumption and boost their memory and processing capability is spintronics [18]. These devices make

use of electrons' and/or holes' spin degrees of freedom, which can also interact with their orbital moments. Spin-orbit coupling or magnetic layers acting as spin-polarisers or analyzers regulate the spin polarization in these devices. Spin current can also be carried by spin waves. Spintronic devices use the spin degree of freedom of electrons. Next-generation memory and logic devices can use MRAM and spin valves because they are fast and don't lose their data [18].

4.2 Optoelectronic Devices

Semiconductor devices that can transform light energy into electrical energy and vice versa are known as optoelectronic devices. This device is constructed from solid crystalline crystals, which are heavier than insulators but lighter than metals. An optoelectronic device is any electrical device that uses light. This technology is employed in a variety of optoelectronics applications, such as military, telecommunications, automatic access control systems, and medical devices [19-22].

5. Challenges and Limitations

The field of advanced electronic materials and emerging device technologies has made significant progress; however, it still faces numerous challenges. One of the most pressing issues is the need to develop high-quality materials that exhibit predictable properties and consistent performance. Thus, the long-term stability and environmental endurance of organic semiconductors and flexible electronic materials are both major challenges. Many organic materials deteriorate in the presence of moisture, oxygen, and high temperatures, limiting their practical use. To overcome these obstacles, we need interdisciplinary collaborations across materials research, device engineering, manufacturing technology, and sustainable resource management.

6. Conclusion

In conclusion, 2D materials with developing structures are essential for electronic devices that go beyond Si. The search for high-performance, low-power, and highly stable electronic and sensor application solutions has prompted a great deal of research into alternative materials, with developing 2D materials showing promise. As alternatives to REE magnets, Mn-based permanent magnets, such as Mn-Al and Mn-Bi, are being thoroughly researched. Magnets made of MnAl and MnBi offer a lot of promise for a variety of moderate-performance uses.

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